

EXP 9

PROGRAM

1. TimeServer.java

```
import java.net.*;

public class TimeServer {
    public static void main(String[] args) throws Exception {

        DatagramSocket socket = new DatagramSocket(5000);
        System.out.println("Time Server is running...");

        while (true) {
            byte[] buffer = new byte[100];

            DatagramPacket request =
                new DatagramPacket(buffer, buffer.length);

            socket.receive(request);

            // Create a new thread for each client
            new Thread(() -> {
                try {
                    String time = new java.util.Date().toString();
                    byte[] data = time.getBytes();

                    DatagramPacket response =
                        new DatagramPacket(
                            data,
                            data.length,
                            request.getAddress(),
                            request.getPort()
                        );

                    socket.send(response);

                } catch (Exception e) {
                    e.printStackTrace();
                }
            }).start();
        }
    }
}
```

```
    }  
  }  
}
```

2. TimeClient.java

```
import java.net.*;  
  
public class TimeClient {  
    public static void main(String[] args) throws Exception {  
  
        DatagramSocket socket = new DatagramSocket();  
  
        // Send request to server  
        byte[] data = "TIME".getBytes();  
  
        InetAddress server = InetAddress.getByName("localhost");  
  
        DatagramPacket request =  
            new DatagramPacket(data, data.length, server, 5000);  
  
        socket.send(request);  
  
        // Receive time from server  
        byte[] buffer = new byte[100];  
  
        DatagramPacket response =  
            new DatagramPacket(buffer, buffer.length);  
  
        socket.receive(response);  
  
        String time = new String(  
            response.getData(), 0, response.getLength()  
        );  
  
        System.out.println("Server Time: " + time);  
  
        socket.close();  
    }  
}
```

```
sjcet@sjcet-H81M-H-M-2:~$ gedit TimeServer.java
sjcet@sjcet-H81M-H-M-2:~$ javac TimeServer.java
sjcet@sjcet-H81M-H-M-2:~$ gedit TimeClient.java
sjcet@sjcet-H81M-H-M-2:~$ javac TimeClient.java
sjcet@sjcet-H81M-H-M-2:~$ java TimeServer
Time Server is running...
```

```
sjcet@sjcet-H81M-H-M-2:~$ java TimeClient
Server Time: Thu Oct 08 11:44:05 IST 2026
sjcet@sjcet-H81M-H-M-2:~$
```